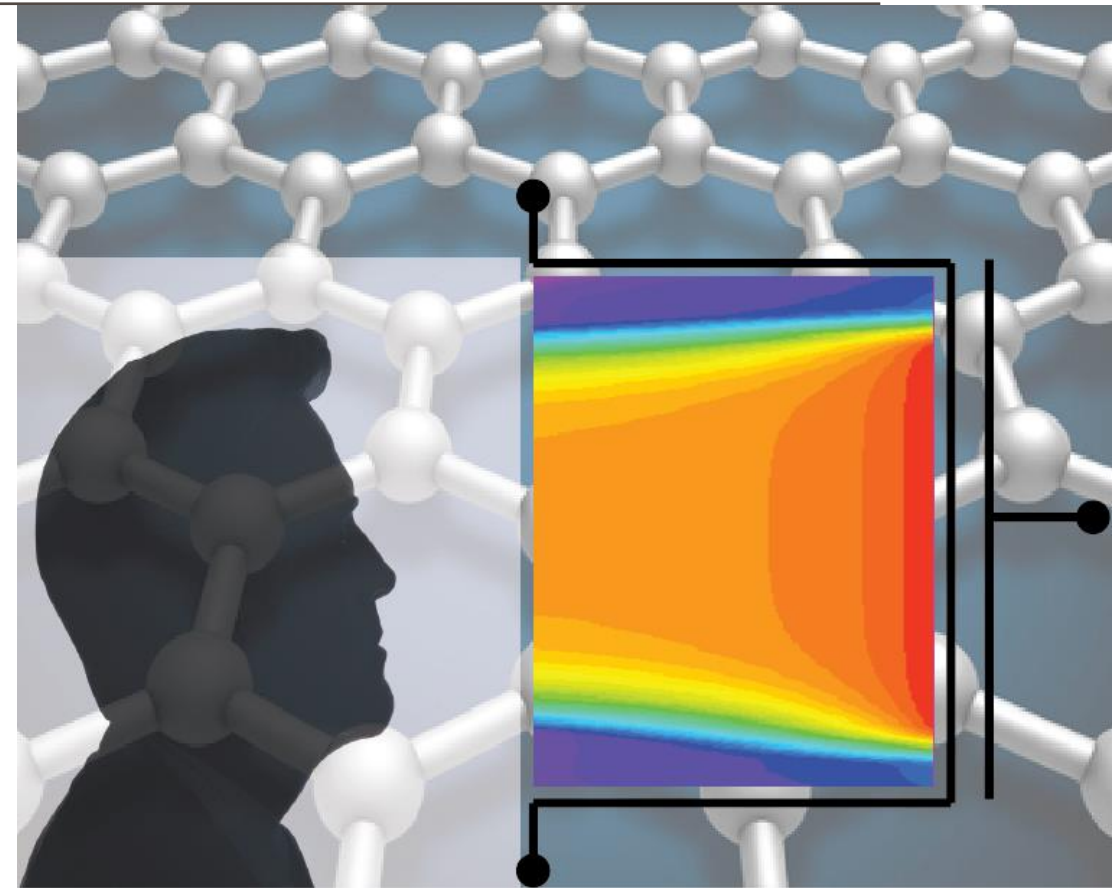


ADVANCED SOLID-STATE DEVICES

Week-4, Lecture-1

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Non-equilibrium Excess Carriers

Rate of change of carrier concentration

$$\frac{dn}{dt} = -\alpha_r \delta n [(n_o + p_o) + \delta n]$$

- This equation is difficult to solve for the general case.
- But can be solved easily when we assume *low-level injection*

Low-Level Injection and High-Level Injections

- Low-level injection: When the excess carrier concentration is small compared to equilibrium majority carrier concentration
- High-level injection: When the excess carrier concentration is comparable or exceeds equilibrium majority carrier concentration

Low-level injection

- For p-type material:
 $(\delta p = \delta n) \ll p_o$
- For n-type material:
 $(\delta p = \delta n) \ll n_o$

Consider n-type semiconductor with $N_D = 1 \times 10^{15} \text{ cm}^{-3}$. Assume $\delta p = \delta n = 1 \times 10^{12} \text{ cm}^{-3}$. Compute n and p . Is it low-level injection?

Answer: $n \approx 1 \times 10^{15} \text{ cm}^{-3}$ and $p = 1 \times 10^{12} \text{ cm}^{-3}$. Yes.

Consider n-type semiconductor with $N_D = 1 \times 10^{15} \text{ cm}^{-3}$. Assume $\delta p = \delta n = 1 \times 10^{17} \text{ cm}^{-3}$. Compute n and p . Is it low-level injection?

Answer: $n = 1.01 \times 10^{17} \text{ cm}^{-3}$ and $p = 1 \times 10^{17} \text{ cm}^{-3}$. No.

Non-equilibrium Excess Carriers

Rate of change of carrier concentration (Low-level Injection)

$$\frac{dn}{dt} = -\alpha_r \delta n [(n_o + p_o) + \delta n]$$

Problem: Assume p-type extrinsic material with low level injection.

- Simplify the above equation.
- Write an expression for electron concentration as a function of time.

p-type extrinsic material: $p_o \gg n_o$

Low-level injection: $p_o \gg \delta n$

$$(n_o + p_o) + \delta n \approx p_o$$

$$\frac{dn}{dt} = -\alpha_r p_o \delta n$$

$$\frac{d(n_o + \delta n)}{dt} = -\alpha_r p_o \delta n$$

$$\frac{d(\delta n)}{dt} = -\alpha_r p_o \delta n$$

$$\delta n = c e^{-\alpha_r p_o t} \quad [c \text{ is } \delta n \text{ at } t = 0]$$

$$n = n_o + c e^{-\alpha_r p_o t}$$

- We can define: $\tau_{no} = \frac{1}{\alpha_r p_o}$. Hence: $\delta n = c e^{-t/\tau_{no}}$
- τ_{no} describes time of decay of excess carriers
- We call it *excess minority carrier lifetime (of electrons)*

Recombination Rates

The recombination rate is defined as $-\frac{d(\delta n)}{dt}$

Hence, recombination rate of electrons (minority carriers in p-type) is:

$$R'_n = \frac{\delta n}{\tau_{no}}$$

The recombination rate of the holes (majority carrier in p-type) is same:

$$R'_p = \frac{\delta p}{\tau_{po}}$$

- The lifetime of the electrons (minority carriers) primarily governs the recombination rate since they interact with the abundant holes.

We can consider n-type material and define excess minority carrier lifetime for holes τ_{po} :

$$\frac{d(\delta p)}{dt} = -\frac{\delta p}{\tau_{po}}$$

N-type material

Recombination rate of holes (minority carriers in n-type) is:

$$R'_p = \frac{\delta n}{\tau_{po}} \text{ [Note: } \delta n = \delta p \text{]}$$

The recombination rate of the electrons (majority carrier in n-type) is same:

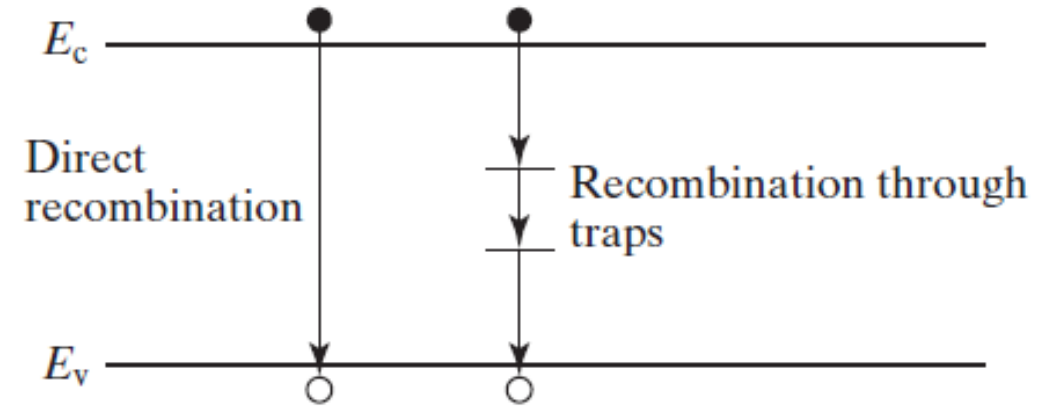
$$R'_n = \frac{\delta n}{\tau_{po}}$$

- The lifetime of the holes (minority carriers) primarily governs the recombination rate since they are the limiting factor

Mechanism of Recombination

Recombination can take place using two kinds of process:

1. **Direct Band-to-Band Recombination:** Electron drops directly from conduction band to an unoccupied state in the valence band
2. **Indirect Recombination:** Electron first makes a transition to an energy level in the band gap and then it captures hole from the valence band



Recombination: Energy Release

- After recombination, the energy difference between electron and hole energy is released.

- The energy can release in two major ways:
 1. Radiative: energy is released as photon
 2. Non-Radiative: energy is released in some other form

- Non-radiative recombination:
 1. Phonon: energy is released as phonon (lattice)
 2. Augur: energy is given to some other mobile carrier as kinetic energy

Direct Band-to-band Recombination

For direct bandgap semiconductor:

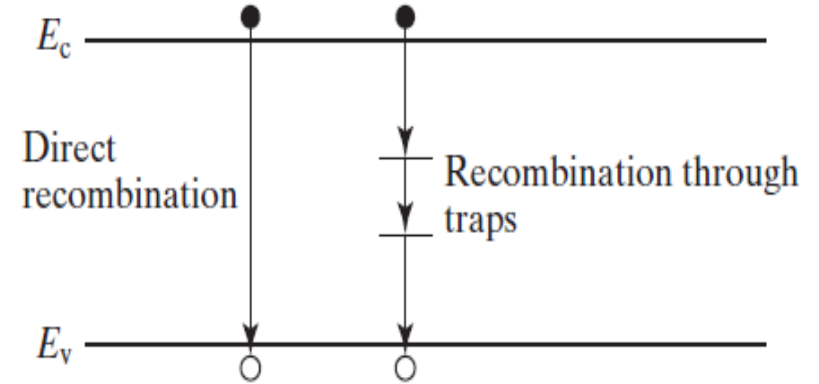
- Energy involved in the direct band-to-band recombination is released as photon (radiative recombination)
- Example: GaAs

For indirect bandgap semiconductor:

- Involves change in momentum (requires emission or absorption of phonons)
- Very low probability of direct band-to-band recombination in an indirect bandgap semiconductor
- Example: Ge, Si very low probability of direct band-to-band recombination

Indirect Recombination

- Indirect recombination can occur with the help of trap energy level
- How are these energy levels created?
 - Impurity atoms other than donors and acceptors
 - Crystal defects
- Why are the trap energy levels important?
 - Can greatly enhance the recombination by acting as an intermediate energy level



Steps of Recombination using Traps (1)

Four basic steps:

1. Electron capture
2. Electron emission
3. Hole capture
4. Hole emission

- Shockley Read Hall Theory of Recombination
- SRH Recombination

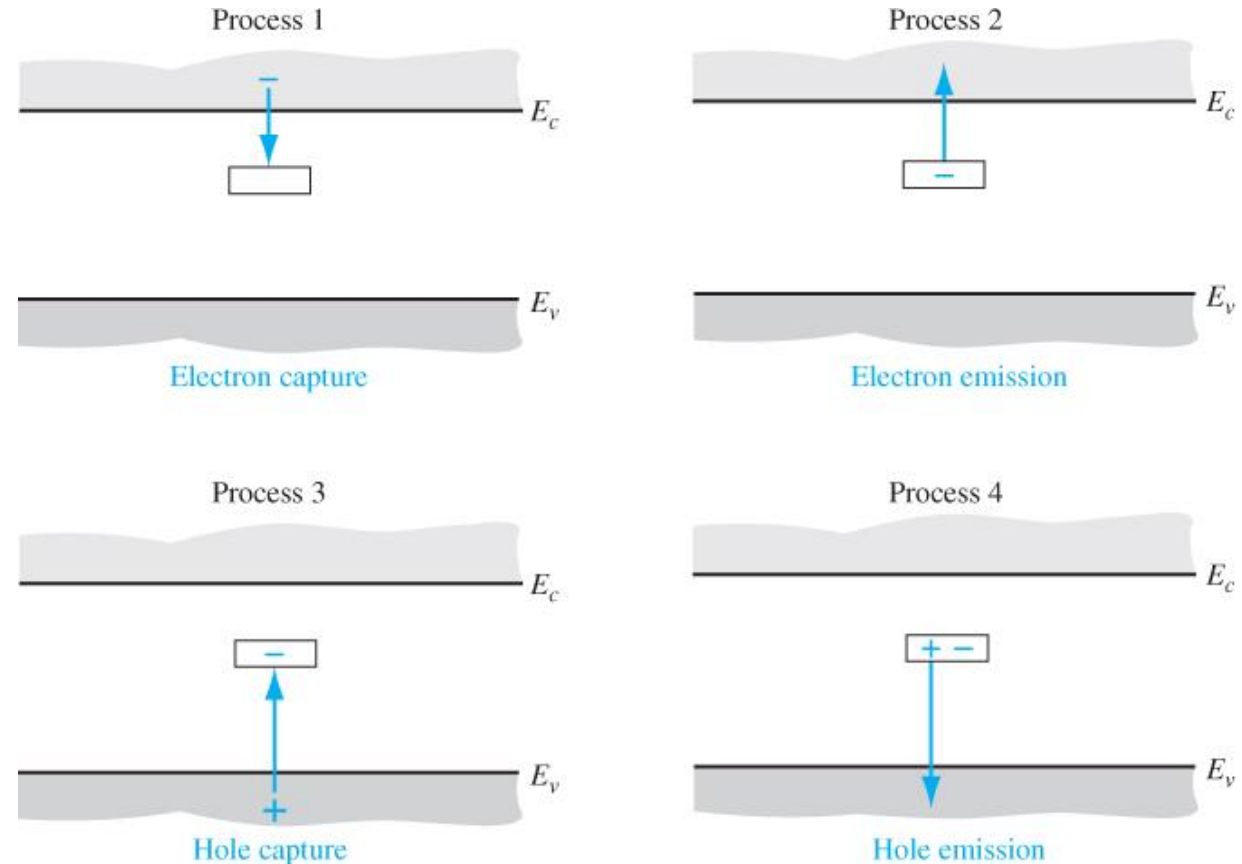


Figure 6.16 | The four basic trapping and emission processes for the case of an acceptor-type trap.

SRH Recombination Theory

- Neutrality conditions:
 - Excess electrons and holes are equal
 - Recombination rates of electrons and holes are equal, i.e., $R_n = R_p$

- The driving force for recombination is the difference between np and n_i^2 .
- In equilibrium no recombination

$$R_n = R_p = \frac{(np - n_i^2)}{\tau_{po}(n + n') + \tau_{no}(p + p')}$$

- Carrier lifetimes τ_{no} and τ_{po}
- $n' = N_C \exp\left(-\frac{E_C - E_t}{kT}\right)$
- $p' = N_V \exp\left(-\frac{E_t - E_V}{kT}\right)$

Capture and Emission: Extrinsic and Low-level Injection

$$R_n = R_p = \frac{(np - n_i^2)}{\tau_{po}(n + n') + \tau_{no}(p + p')}$$

Problem: Assume that the material is extrinsic n-type and is having low-level excess carrier injection.

Further assume that the trap states are near the middle of band gap. Simplify the above expression.

Solution: $n_0 \gg p_0$ and $n_0 \gg (\delta n = \delta p)$

The trap level is somewhere in the middle, implies that $n > (n' = p')$

$$\begin{aligned} R_n = R_p &= \frac{(np - n_i^2)}{\tau_{po}n} \\ &= \frac{(n_o + \delta n)(p_o + \delta n) - n_i^2}{\tau_{po}(n_o + \delta n)} = \frac{n_o \cdot \delta n}{\tau_{po}n_o} \\ R_n = R_p &= \frac{\delta n}{\tau_{po}} \end{aligned}$$

- The recombination rate is decided by the excess minority carrier lifetime
- When lifetime of carrier reduces, the recombination becomes faster

Similarly, we can derive for p-type material: $R_n = R_p = \frac{\delta n}{\tau_{no}}$ [depends on lifetime of electrons]

Capture and Emission: Intrinsic and Low-level Injection

$$R_n = R_p = \frac{(np - n_i^2)}{\tau_{po}(n + n') + \tau_{no}(p + p')}$$

Problem: Assume that the material is intrinsic and low level injection. Given: $n = n_i + \delta n$ and $p = n_i + \delta n$.

Also assume, $n' = p' = n_i$

Find the rate of recombination.

Answer: $R_n = R_p = \frac{\delta n}{\tau_{no} + \tau_{po}}$

Note that for extrinsic semiconductor we had derived:

for n-type material: $R_n = R_p = \frac{\delta p}{\tau_{po}}$
[depends on lifetime of holes]

for p-type material: $R_n = R_p = \frac{\delta n}{\tau_{no}}$
[depends on lifetime of electrons]

- The effective excess carrier lifetime of carrier **increases** in intrinsic semiconductor.
- Number of majority carriers available for the minority carriers to recombine reduces in an intrinsic semiconductor

Non-equilibrium Excess Carriers



Quasi-Fermi Levels

Quasi-Fermi Levels (1)

For equilibrium case (only one Fermi level):

$$n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right) \quad p_0 = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

- Fermi level is strictly valid for equilibrium condition (no excess carriers are present)

- For non-equilibrium condition, we can write steady state carrier concentration in terms of two *quasi-Fermi levels* (one Fermi level is not sufficient)
- We define quasi-Fermi levels E_{Fn} for electrons and E_{Fp} for holes separately.

Non-equilibrium electron concentration

$$n = n_0 + \delta n = n_i \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right)$$

Non-equilibrium hole concentration:

$$p = p_0 + \delta p = n_i \exp\left(\frac{E_{Fi} - E_{Fp}}{kT}\right)$$

Quasi-Fermi Levels (2)

Non-equilibrium electron concentration

$$n = n_0 + \delta n = n_i \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right)$$

Product of n and p

$$np = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right)$$

Non-equilibrium hole concentration:

$$p = p_0 + \delta p = n_i \exp\left(\frac{E_{Fi} - E_{Fp}}{kT}\right)$$

When $np \neq n_i^2$, (there is a non-equilibrium condition) the two quasi-Fermi levels will be separated

$$np = n_i^2 \exp\left(\frac{E_{Fn} - E_{Fp}}{kT}\right)$$

Quasi-Fermi Levels (3)

Problem: Consider n-type semiconductor, at 300K with $n_0 = 10^{15} \text{ cm}^{-3}$. Assume $n_i = 10^{10} \text{ cm}^{-3}$. a) Compute the position of equilibrium Fermi energy level.

b) Assume that there is an excess charge $\delta n = \delta p = 10^{13} \text{ cm}^{-3}$. Compute the position of quasi-Fermi level for electron and holes.

$$a) \quad n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right)$$

$$E_F - E_{Fi} = kT \ln\left(\frac{n_0}{n_i}\right) = 0.2982 \text{ eV}$$

$$b) \quad n = n_0 + \delta n = 1.01 \times 10^{15} =$$

$$n_i \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right)$$

$$E_{Fn} - E_{Fi} = 0.2984 \text{ eV}$$

$$p = p_0 + \delta p = 10^{13} = n_i \exp\left(\frac{E_{Fi} - E_{Fp}}{kT}\right)$$

$$E_{Fi} - E_{Fp} = 0.179 \text{ eV}$$

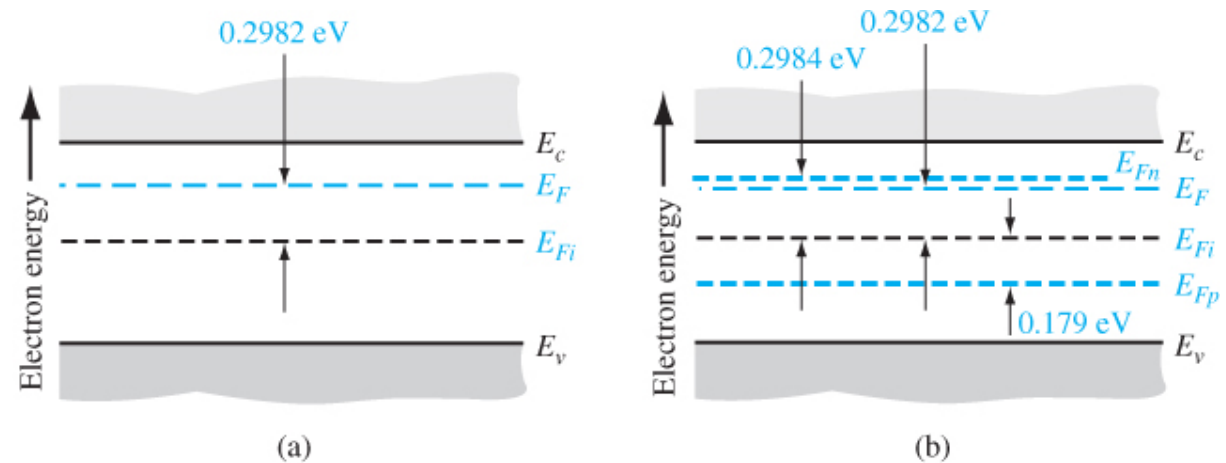


Figure 6.15 | (a) Thermal-equilibrium energy-band diagram for $N_d = 10^{15} \text{ cm}^{-3}$ and $n_i = 10^{10} \text{ cm}^{-3}$. (b) Quasi-Fermi levels for electrons and holes if 10^{13} cm^{-3} excess carriers are present.

Quasi-Fermi Levels: Significance

- Analyzing devices in non-equilibrium condition, becomes easy using *quasi-Fermi levels*

- The gradient of quasi-Fermi levels determines the current density
- Often it is easy to compute the gradients of the quasi-Fermi levels

- Current density:

$$J = J_n + J_p = qn\mu_n E_x + qD_n \frac{dn}{dx} + qp\mu_p E_x - qD_p \frac{dp}{dx}$$

We have discussed electric field can be written:

$$E(x) = \frac{1}{q} \frac{dE_c(x)}{dx} = \frac{1}{q} \frac{dE_{Fi}(x)}{dx}$$

Non-equilibrium electron concentration

$$n = n_i \exp\left(\frac{E_{Fn} - E_{Fi}}{kT}\right)$$

- Current density in terms of quasi-Fermi level (can be easily proven):

$$J_n = \mu_n n \frac{dE_{Fn}}{dx}$$

$$J_p = \mu_p p \frac{dE_{Fp}}{dx}$$

- Both drift and diffusion components are accounted for in the above equations

- Often it is easy to compute the gradients of the quasi-Fermi levels

References

- D. A. Neaman, "*Semiconductor Physics and Devices*"